Report
Full Sales Funnel Analysis
(User Funnel)

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1. Introduction & Goals

Funnel Analysis is one of the key methods in product analytics. It tracks the user journey from the first interaction with a product to a target action (purchase, registration, etc.) and identifies stages with the highest drop-off rates.

The dataset contains information on 17,175 user events across 5 stages of an e-commerce funnel:

Stage	Description	Users
homepage	Home page — first visit to the website	10,000
product_page	Product page — viewing a product card	5,000
cart	Shopping cart — adding a product to cart	1,500
checkout	Checkout — proceeding to payment	450
purchase	Purchase — successful transaction completed	225

2. Data Acquisition & Loading

2.1 Data Source

The dataset is sourced from Kaggle: "User Funnels Dataset". The file user_data.csv contains three columns:

Column	Type	Description
user_id	STRING	Unique user identifier (user_0 … user_N)
stage	STRING	Funnel stage: homepage, product_page, cart, checkout, purchase
conversion	BOOLEAN	True = user proceeded to the next stage; False = dropped off

2.2 Creating the Table in SQL

First, create the table in the database (compatible with PostgreSQL / BigQuery / SQLite):

```
-- Create the main funnel table
CREATE TABLE user_funnels (
    user_id    VARCHAR(50)  NOT NULL,
    stage      VARCHAR(50)  NOT NULL,
    conversion BOOLEAN      NOT NULL
);

-- Load data from CSV (PostgreSQL)
COPY user_funnels(user_id, stage, conversion)
FROM '/path/to/user_data.csv'
DELIMITER ','
CSV HEADER;
```

3. Data Cleaning & Validation

3.1 Check for NULL and Empty Values

```
-- Find NULL values in all columns
SELECT
    COUNT(*) FILTER (WHERE user_id IS NULL)    AS null_user_id,
    COUNT(*) FILTER (WHERE stage IS NULL)      AS null_stage,
    COUNT(*) FILTER (WHERE conversion IS NULL) AS null_conversion,
    COUNT(*)                                   AS total_rows
FROM user_funnels;

-- Result: null_user_id=0, null_stage=0, null_conversion=0, total_rows=17175
-- Conclusion: dataset is complete, no missing values
```

3.2 Validate Allowed Values

```
-- Check valid values in the 'stage' field
SELECT stage, COUNT(*) AS cnt
FROM user_funnels
GROUP BY stage
ORDER BY cnt DESC;

-- Expected values: homepage, product_page, cart, checkout, purchase
-- All 5 values are valid, no unexpected entries found

-- Check conversion values
SELECT DISTINCT conversion FROM user_funnels;
-- Result: True, False — OK
```

3.3 Check for Duplicates

```
-- Find duplicate records (same user at the same stage)
SELECT user_id, stage, COUNT(*) AS cnt
FROM user_funnels
GROUP BY user_id, stage
HAVING COUNT(*) > 1;

-- Empty result = no duplicates found
```

3.4 Cleaning Summary

The dataset requires no additional cleaning: there are no NULL values, unexpected categories, or duplicate records. Data is ready for analysis.

4. Exploratory Data Analysis (EDA)

4.1 General Statistics

```
-- Total records and unique users
SELECT
    COUNT(*)                AS total_events,
    COUNT(DISTINCT user_id) AS unique_users
FROM user_funnels;

-- Distribution by stage
SELECT
    stage,
    COUNT(*) AS user_count,
    ROUND(COUNT(*) * 100.0 / SUM(COUNT(*)) OVER (), 2) AS pct_of_total
FROM user_funnels
GROUP BY stage
ORDER BY user_count DESC;
```

Stage	Users	% of Total Records
homepage	10,000	58.2%
product_page	5,000	29.1%
cart	1,500	8.7%
checkout	450	2.6%
purchase	225	1.3%

4.2 Conversion Field Distribution

```
-- True / False breakdown
SELECT
    conversion,
    COUNT(*) AS cnt,
    ROUND(COUNT(*) * 100.0 / SUM(COUNT(*)) OVER (), 1) AS pct
FROM user_funnels
GROUP BY conversion;

-- conversion = True:  13,014 (75.8%)
-- conversion = False:  4,161 (24.2%)
```

5. Sales Funnel Analysis

5.1 Users at Each Funnel Stage

```
-- Number of users at each step of the funnel
SELECT
    stage,
    COUNT(*) AS users_at_stage
FROM user_funnels
GROUP BY stage
ORDER BY
    CASE stage
        WHEN 'homepage'     THEN 1
        WHEN 'product_page' THEN 2
        WHEN 'cart'         THEN 3
        WHEN 'checkout'     THEN 4
        WHEN 'purchase'     THEN 5
    END;
```

5.2 Step-to-Step Conversion Rate

The core funnel metric — conversion from the previous step. Shows what percentage of users proceeded to the next stage.

```
-- Step-to-step conversion between adjacent stages
WITH stage_counts AS (
    SELECT
        stage,
        COUNT(*) AS cnt,
        CASE stage
            WHEN 'homepage'     THEN 1
            WHEN 'product_page' THEN 2
            WHEN 'cart'         THEN 3
            WHEN 'checkout'     THEN 4
            WHEN 'purchase'     THEN 5
        END AS stage_order
    FROM user_funnels
    GROUP BY stage
```

```
),
with_prev AS (
    SELECT
        stage, cnt, stage_order,
        LAG(cnt) OVER (ORDER BY stage_order) AS prev_cnt
    FROM stage_counts
)
SELECT
    stage,
    cnt,
    prev_cnt,
    ROUND(cnt * 100.0 / NULLIF(prev_cnt, 0), 1) AS step_conversion_pct
FROM with_prev
ORDER BY stage_order;
```

Stage	Users	From Previous Stage	Step Conversion
homepage	10,000	—	—
product_page	5,000	10,000	50.0% ↘
cart	1,500	5,000	30.0% ↘↘
checkout	450	1,500	30.0% ↘↘
purchase	225	450	50.0% ↘

Critical bottlenecks: product_page → cart (−70%) and cart → checkout (−70%). These two steps account for the largest user losses.

5.3 Overall Conversion Rate (Top to Bottom)

```
-- Overall conversion rate: homepage → purchase
WITH top AS (
    SELECT COUNT(*) AS total FROM user_funnels WHERE stage = 'homepage'
),
bottom AS (
    SELECT COUNT(*) AS purchased FROM user_funnels WHERE stage = 'purchase'
)
SELECT
    total,
    purchased,
    ROUND(purchased * 100.0 / total, 2) AS overall_conversion_pct
FROM top, bottom;

-- Result: total=10,000, purchased=225, overall_conversion_pct=2.25%
```

5.4 Conversion Using the 'conversion' Field

```
-- Conversion rate per stage based on the conversion column
SELECT
    stage,
    COUNT(*) AS total_users,
    COUNT(*) FILTER (WHERE conversion = TRUE)  AS converted,
    COUNT(*) FILTER (WHERE conversion = FALSE) AS dropped,
    ROUND(
        COUNT(*) FILTER (WHERE conversion = TRUE) * 100.0 / COUNT(*),
    1) AS conversion_rate_pct
FROM user_funnels
GROUP BY stage
ORDER BY
    CASE stage
        WHEN 'homepage'     THEN 1
        WHEN 'product_page' THEN 2
        WHEN 'cart'         THEN 3
        WHEN 'checkout'     THEN 4
        WHEN 'purchase'     THEN 5
    END;
```

5.5 Drop-off Rate Analysis

```
-- User drop-off at each stage
WITH stage_counts AS (
    SELECT
        stage, COUNT(*) AS cnt,
        CASE stage
            WHEN 'homepage'     THEN 1
            WHEN 'product_page' THEN 2
            WHEN 'cart'         THEN 3
            WHEN 'checkout'     THEN 4
            WHEN 'purchase'     THEN 5
        END AS ord
    FROM user_funnels GROUP BY stage
),
with_next AS (
    SELECT stage, cnt, ord,
        LEAD(cnt) OVER (ORDER BY ord) AS next_cnt
    FROM stage_counts
)
SELECT
    stage,
    cnt,
    cnt - COALESCE(next_cnt, cnt)                           AS dropped_users,
    ROUND((cnt - COALESCE(next_cnt, cnt)) * 100.0 / cnt, 1) AS dropoff_pct
FROM with_next
ORDER BY ord;
```

Stage	Users	Dropped (count)	Drop-off %
homepage	10,000	5,000	50.0%
product_page	5,000	3,500	70.0%
cart	1,500	1,050	70.0%
checkout	450	225	50.0%
purchase	225	0	—

5.6 Cumulative Conversion from the Top of Funnel

```
-- Conversion of each stage relative to homepage (top-of-funnel)
WITH top AS (
    SELECT COUNT(*) AS top_cnt FROM user_funnels WHERE stage = 'homepage'
)
SELECT
    uf.stage,
    COUNT(*) AS stage_users,
    top.top_cnt,
    ROUND(COUNT(*) * 100.0 / top.top_cnt, 2) AS cumulative_conversion_pct
FROM user_funnels uf, top
GROUP BY uf.stage, top.top_cnt
ORDER BY
    CASE uf.stage
        WHEN 'homepage'     THEN 1
        WHEN 'product_page' THEN 2
        WHEN 'cart'         THEN 3
        WHEN 'checkout'     THEN 4
        WHEN 'purchase'     THEN 5
    END;
```

Stage	Users	Cumulative Conversion %
homepage	10,000	100.00%
product_page	5,000	50.00%
cart	1,500	15.00%
checkout	450	4.50%
purchase	225	2.25%

6. User Segmentation

6.1 Classify Users by Funnel Depth

```
-- Determine the deepest stage reached by each user
SELECT
    user_id,
    MAX(CASE stage
        WHEN 'homepage'     THEN 1
        WHEN 'product_page' THEN 2
        WHEN 'cart'         THEN 3
        WHEN 'checkout'     THEN 4
        WHEN 'purchase'     THEN 5
    END) AS max_stage_order
FROM user_funnels
GROUP BY user_id;

-- Group by user segment
WITH user_depth AS (
    SELECT user_id,
        MAX(CASE stage
            WHEN 'homepage'     THEN 1
            WHEN 'product_page' THEN 2
            WHEN 'cart'         THEN 3
            WHEN 'checkout'     THEN 4
            WHEN 'purchase'     THEN 5
        END) AS max_depth
    FROM user_funnels GROUP BY user_id
)
SELECT
    CASE max_depth
        WHEN 1 THEN 'Bounce (homepage only)'
        WHEN 2 THEN 'Browser (reached product page)'
        WHEN 3 THEN 'Intent (reached cart)'
        WHEN 4 THEN 'Near-Buyer (reached checkout)'
        WHEN 5 THEN 'Buyer (completed purchase)'
    END AS user_segment,
    COUNT(*) AS user_count,
    ROUND(COUNT(*) * 100.0 / SUM(COUNT(*)) OVER (), 2) AS pct
FROM user_depth
GROUP BY max_depth
ORDER BY max_depth;
```

6.2 Cart Abandonment — Users Who Dropped at Cart

```
-- Users who reached cart but did not proceed to checkout
SELECT DISTINCT uf.user_id
FROM user_funnels uf
WHERE uf.stage = 'cart'
  AND uf.user_id NOT IN (
      SELECT user_id FROM user_funnels WHERE stage = 'checkout'
  );
```

```
-- Total count of cart abandoners
SELECT COUNT(DISTINCT user_id) AS cart_abandoners
FROM user_funnels
WHERE stage = 'cart'
  AND user_id NOT IN (
      SELECT user_id FROM user_funnels WHERE stage = 'checkout'
  );
-- Result: 1,050 users abandoned their cart
```

6.3 Checkout Abandonment

```
-- Users who started checkout but did not complete the purchase
SELECT COUNT(DISTINCT user_id) AS checkout_abandoners
FROM user_funnels
WHERE stage = 'checkout'
  AND user_id NOT IN (
      SELECT user_id FROM user_funnels WHERE stage = 'purchase'
  );
-- Result: 225 users
```

7. Advanced SQL Metrics

7.1 Funnel Using Window Functions

```
-- Full conversion metrics using window functions (PostgreSQL / BigQuery)
WITH stage_counts AS (
    SELECT
        stage,
        COUNT(*) AS cnt,
        CASE stage
            WHEN 'homepage'     THEN 1
            WHEN 'product_page' THEN 2
            WHEN 'cart'         THEN 3
            WHEN 'checkout'     THEN 4
            WHEN 'purchase'     THEN 5
        END AS ord
    FROM user_funnels
    GROUP BY stage
)
SELECT
    stage,
    cnt,
    LAG(cnt)  OVER (ORDER BY ord)  AS prev_stage_cnt,
    LEAD(cnt) OVER (ORDER BY ord)  AS next_stage_cnt,
    FIRST_VALUE(cnt) OVER (ORDER BY ord) AS top_funnel_cnt,
    ROUND(cnt * 100.0 / FIRST_VALUE(cnt) OVER (ORDER BY ord), 2) AS pct_of_top,
    ROUND(cnt * 100.0 / NULLIF(LAG(cnt) OVER (ORDER BY ord), 0), 2) AS step_conv
FROM stage_counts
ORDER BY ord;
```

7.2 Cumulative Drop-off

```
-- Cumulative drop-off (total users lost by a given stage)
WITH stage_counts AS (
    SELECT stage, COUNT(*) AS cnt,
        CASE stage
            WHEN 'homepage'     THEN 1
            WHEN 'product_page' THEN 2
            WHEN 'cart'         THEN 3
            WHEN 'checkout'     THEN 4
            WHEN 'purchase'     THEN 5
        END AS ord
    FROM user_funnels GROUP BY stage
),
top AS (SELECT cnt AS top_cnt FROM stage_counts WHERE ord = 1)
SELECT
    sc.stage,
    sc.cnt,
    top.top_cnt - sc.cnt                                        AS total_lost,
    ROUND((top.top_cnt - sc.cnt) * 100.0 / top.top_cnt, 2)   AS pct_lost_from_top
FROM stage_counts sc, top
ORDER BY sc.ord;
```

8. Key Findings & Recommendations

8.1 Funnel Summary Table

Stage	Users	Step Conv.	Overall Conv.	Drop-off %
homepage	10,000	—	100%	50.0%
product_page	5,000	50.0%	50.0%	70.0%
cart	1,500	30.0%	15.0%	70.0%
checkout	450	30.0%	4.5%	50.0%
purchase	225	50.0%	2.25%	—

8.2 Critical Bottlenecks

Two critical bottlenecks were identified based on the analysis:

1. product_page → cart (30% conversion): 70% of users who viewed a product did not add it to the cart. Possible causes: price, insufficient product information, lack of reviews, poor navigation.

2. cart → checkout (30% conversion): 70% of users abandoned the cart. Causes: unexpected additional costs (shipping), complex checkout process, lack of convenient payment options, forced registration.

8.3 Recommendations

Problem	Recommendation	Expected Impact
product_page → cart: 70% drop	Improve product card UX: add reviews, Q&A, video, clear CTA	+10–15% conversion
product_page → cart: 70% drop	A/B test the Add to Cart button: color, copy, placement	+5–8%
cart → checkout: 70% drop	Show shipping cost upfront; add free shipping above a threshold	+8–12%
cart → checkout: 70% drop	Simplify checkout: allow guest payment without registration	+10–15%
checkout → purchase: 50% drop	Add trust signals: SSL badges, guarantees, return policy	+5–7%
Overall CR 2.25%	Email retargeting for cart abandoners with personalized offers	+20% of lost users